



A Hybrid CAT-GAN and ANFIS-based Multimodal Framework for Explainable Depth of Anesthesia Classification



Abhishek Gupta

M.Sc. Computer Science (Big Data Analytics)

Under the Guidance of

Dr. Subodh Kumar

Assistant Professor

Department of Data Science and Analytics

Central University of Rajasthan, Ajmer

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Introduction - The Clinical Challenge

- Anesthesia is a complex pharmacological triad of **hypnosis, analgesia, and areflexia (immobility)** .
- **BIS (Bispectral Index)**[0-100] is the current commercial gold standard for tracking this depth.
- **The Data Gap:** Raw EEG signals are *high-dimensional and chaotic*, making real-time manual interpretation impossible during surgery.
- **Vulnerability:** Commercial monitors rely purely on single-channel EEG, making them highly susceptible to electrical interference and physical detachment in active operating rooms.



Figure 1: Anesthesia Workstation.

The Resource Reality & The Fuzzy Advantage

Current Limitations in Indian Healthcare

- **The Vigilance Gap:** In Tier-2/3 healthcare, a scarcity of dedicated anesthesiologists leads to "Serial Monitoring," significantly increasing the risk of intraoperative awareness.
- **Lagging Indicators:** Autonomic vitals (HR, BP) react *after* the patient begins to wake up.
- **The Black-Box Crisis:** Modern deep learning architectures offer high accuracy but provide zero physiological transparency, leading to clinical distrust.

The Logic of Fuzzy Systems & ANFIS

- **Handling Ambiguity:** Safely maps the non-linear "Grey Areas" of biological transitions using continuous membership gradients.
- **Explainable AI (XAI):** Unlike neural networks, ANFIS extracts a transparent *If – Then* Sugeno rule base that clinicians can physically audit and verify [10].
- **Multimodal Fusion:** Integrates both neurological features and cardiovascular vitals, creating a fail-safe against hardware disruption.

"Moving from reactive alarms to proactive, transparent clinical decision support."

Objectives

- **To engineer a multimodal data fusion framework** capable of integrating non-stationary Electroencephalogram (EEG) signals with synchronized physiological telemetry (vital signs).
- **To extract and compress neurological features** using 4-level Discrete Wavelet Transforms (DWT) alongside different entropies.
- **To resolve severe clinical class imbalances** natively found in surgical datasets by evaluating and applying deep generative manifold algorithms.
- **To design and evaluate an Adaptive Neuro-Fuzzy Inference System (ANFIS)** initialized via Fuzzy C-Means (FCM) to predict depth of anesthesia index with complete clinical interpretability.

Literature Review: Feature Engineering to Deep Learning

- **Güler & Übeyli (2005) [2]:** Established the foundational efficacy of combining Discrete Wavelet Transforms (DWT) with Adaptive Neuro-Fuzzy systems, extracting statistical sub-band markers to achieve 98.68% accuracy across discrete states .
- **Shi et al. (2023) [8]:** Highlighted the transition to deep learning by proposing a **Deep Residual Shrinkage Network (DRSN)** to handle clinical noise. Achieved a accuracy of 95.03% but functioned as a strict "black box" .
- **Zhan et al. (2021) [3]:** Evaluated the efficacy of **Heart Rate Variability (HRV)** as a primary marker for anesthesia state classification. By utilizing a Deep Neural Network (DNN) and time-domain features, the study demonstrated that autonomic nervous system responses can independently distinguish sedation levels with 90.1% accuracy .
- **Tu et al. (2024) [5]:** Evaluated machine learning-based monitoring using single-channel EEG recording. Achieved ~90% accuracy but highlighted the fragility of single-modality systems when faced with intraoperative signal artifacts.

Literature Review: Interpretable AI & Data Balancing

- **Shalbaf et al. (2018) [6]:** Addressed the transparency crisis by engineering an EEG-based **ANFIS-LH**. It utilized explicitly selected spectral and fractal features to classify states, mimicking human cognitive reasoning with 92% accuracy.
- **Sharma, Neeraj et al. (2024) [1]:** Developed **XAI-VSDoA**, an XGBoost-based scheme using exclusively **Vital Signs**. Demonstrated 99.34% accuracy and utilized SHAP and LIME to prove that autonomic markers (Heart Rate, SpO2) are statistically significant proxies for BIS.
- **The Class Imbalance Problem:** Surgical datasets are massively skewed toward stable "Over-Sedation." Linear balancing algorithms often fail to map the true curvature of biological data.
- **Kunakorntum et al. (2020) [7]:** Addressed interpolation noise by introducing **SyMProD**, generating synthetic minority samples based strictly on probabilistic Z-score distributions.
- **Springenberg (2016) [4]:** Introduced the **Categorical Generative Adversarial Network (CAT-GAN)**, utilizing deep generative modeling to map highly complex, non-linear biological manifolds without suffering from mode collapse.

Proposed Methodology

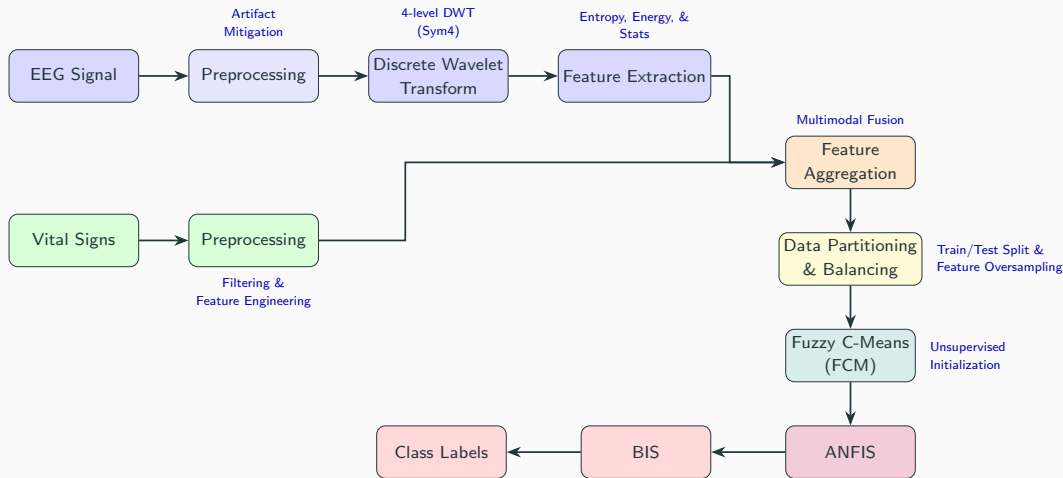
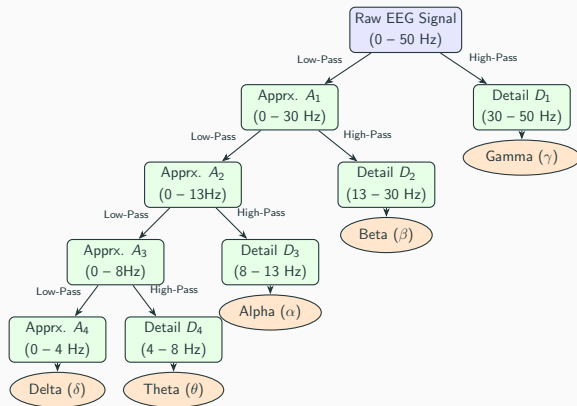


Figure 2: Proposed Multimodal Workflow for Depth of Anesthesia Assessment, incorporating Feature-Space Balancing, FCM, and ANFIS.

EEG Feature Extraction: 4-Level DWT Decomposition



Frequency Mapping

Coeff.	Range	Band
D_1	30–50 Hz	Gamma
D_2	13–30 Hz	Beta
D_3	8–13 Hz	Alpha
D_4	4–8 Hz	Theta
A_4	0–4 Hz	Delta

Key Logic:

- Uses **Symlet-4** wavelet for morphology matching.
- Successive filtering and downsampling.
- Maps non-stationary signals to clinical features.

Figure 3: 4-level DWT decomposition using successive filtering to extract clinically standardized frequency boundaries from non-stationary EEG waveforms.

The Core ANFIS Architecture

The 5-Layer Adaptive Network [10]

- **Layer1(Fuzzification)** Inputs are mapped using the Generalized Bell Membership Function to capture smooth, non-linear physiological transitions.
- **Layer 2 (Rule Firing):** Applies T-norm operators (AND logic) to calculate the firing strength (w_i) of each clinical rule.
- **Layer 3 (Normalization):** Calculates the ratio of the i -th rule's firing strength to the sum of all firing strengths.
- **Layer 4 (Consequent):** Computes the First-Order Takagi-Sugeno polynomial outputs based on the normalized weights.
- **Layer 5 (Defuzzification):** Computes the final, crisp BIS output through an overall weighted summation.

Project-Specific Implementation

- **Initialization:** To prevent the "Curse of Dimensionality" **Fuzzy C-Means (FCM)** clustering was used to intelligently group the data and define the initial rule base.

Generalized Bell Membership Functions (GBMF) defined as:

$$\mu(x) = \frac{1}{1 + \left| \frac{x-c}{a} \right|^{2b}}$$

- a : Determines half-width of curve
- b : slope of curve
- c : center of curve

Dataset Description

1:Data Provenance & Raw Telemetry

The framework utilizes a highly structured clinical dataset sourced from the **University of Queensland** [9]. It initially comprised **12,085 distinct surgical epochs**, synchronizing 2-second continuous EEG windows with cardiovascular vitals.

2:Refinement & Partitioning (80/20 Rule)

Following rigorous artifact removal, preprocessing, and feature aggregation, the refined dataset yielded **11,527 high-fidelity epochs**. To prevent mathematical data leakage during generative balancing, this final dataset is strictly partitioned:

- **80% Training Set:** Isolated strictly for oversampling .
- **20% Test Set:** Sequestered as an unseen hold-out set for rigorous final evaluation.

Early Data Fusion (93 Features)

Rather than processing signals in isolation, the streams are horizontally concatenated per epoch into a **93-dimensional feature space**:

- **Wavelet Statistics:** Mean, Variance, Skewness, and Kurtosis across 5 sub-bands (cA4, cD4-cD1).
- **Non-Linear Dynamics:** Shannon, Sample, Approximate, and Permutation Entropies.
- **Spectral Energy:** Absolute and relative power of clinical frequency bands ($\delta, \theta, \alpha, \beta, \gamma$).
- **Hemodynamics:** Z-score normalized physiological vitals (HR, NBP, etCO₂, etc.).

Target Variables & Clinical Class Labels

The dataset maps the Bispectral Index (BIS) into three distinct macro-labels. This aggregates adjacent physiological states to maximize mathematical separability, but reveals a severe, real-world class imbalance:

Label	Clinical State	Target Index Range	Frequency (Epochs)
Label 0	Transitional / Awake	$BIS \geq 60$	360 (Critical Minority)
Label 1	General Anesthesia	$40 \leq BIS < 60$	2,209 (Optimal Target)
Label 2	Over-Sedation	$BIS < 40$	9,516 (Massive Majority)

The Mathematical Challenge

Because patients spend the vast majority of surgery in Label 2 (Over-Sedation), Architecture suffer from majority-class collapse, failing to detect the critical Label 0 state. This necessitates advanced generative data balancing.

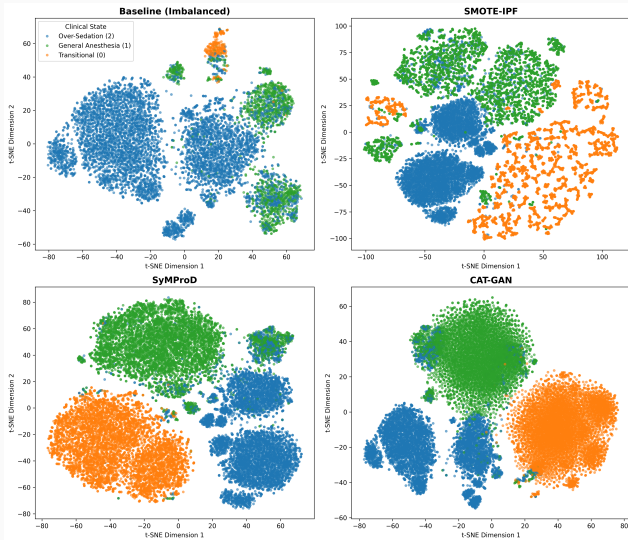
Resolving Clinical Class Imbalance

- Surgical data inherently suffers from majority-class collapse, as patients spend the vast majority of time in stable "Over-Sedation."
- Three balancing techniques were isolated to the training feature space and evaluated via **t-SNE manifold projections**.

Balancing Method	Awake Precision	Awake Recall	Awake F1-Score	Overall Accuracy
Baseline (Imbalanced)	-	-	-	Majority Collapse
SMOTE-IPF (Linear)	69.0%	94.0%	0.80	89.98%
SyMProD (Parametric)	89.0%	83.0%	0.86	90.07%
CAT-GAN (Deep Generative)	92.0%	79.0%	0.85	92.19%

- **Conclusion:** SMOTE-IPF created noisy linear boundaries leading to false alarms (low precision). CAT-GAN successfully mapped the non-linear biological manifold, yielding the safest operational threshold and highest overall accuracy.

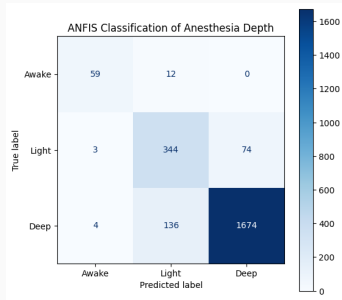
Feature Space Geometry (t-SNE)



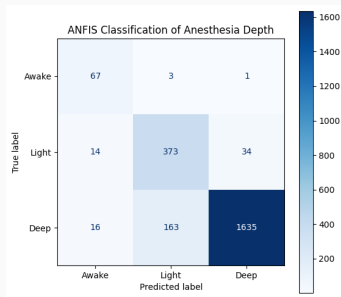
Impact of Balancing:

- **Baseline:** Exhibits severe majority class collapse.
- **SMOTE-IPF:** Shows linear synthetic artifacts, leading to noisy boundaries.
- **SyMProD:** Displays rigid parametric clusters but misses extreme physiological outliers.
- **CAT-GAN (Proposed):** Successfully generates a realistic, non-linear manifold for the minority Transitional state.

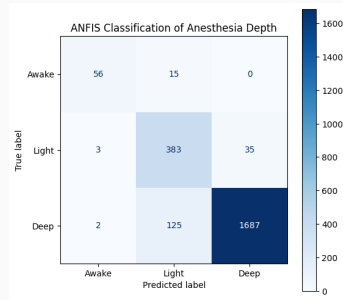
Confusion Matrix Progression



(a) SyMProD



(b) SMOTE-IPF



(c) CAT-GAN (Proposed)

Results: Explainable AI & Clinical Rule Extraction

- Unlike deep neural networks, the optimized ANFIS translates its internal parameters into human-readable **First-Order Sugeno rules**.
- **Feature Relevance:** The network automatically assigned the highest mathematical weight to **Inspired Minimum CO_2 (imCO2)** (0.1580), followed by **Relative Gamma Frequency Energy** (0.0615).

Extracted Clinical Heuristics

Rule 0 (Transitional / Awake State):

IF Inspired CO_2 is extremely LOW ($c = 0.0390$) AND Relative Gamma Energy remains ELEVATED,
THEN patient is Awake (Output Equation pushes BIS towards 80-100).

Rule 1 (Light Anesthesia plane):

IF Inspired CO_2 settles at an INTERMEDIATE plateau ($c = 0.1004$) AND Relative Gamma Energy begins to DECAY,
THEN patient is stable (Output Equation targets mid-range surgical anesthesia: $40 \leq \text{BIS} < 60$).

Rule 2 (Deep Anesthesia):

IF Inspired CO_2 shifts to a HIGHER baseline ($c \approx 0.0851$) AND Relative Gamma Energy is SUPPRESSED,
THEN patient is actively sedated (Output Equation suppresses BIS below 40).

Conclusion & Future Outlook

- **Explainable AI Achieved:** Bypassed the "black-box" interpretability crisis of pure deep learning. The CAT-GAN optimized ANFIS demonstrated textbook-accurate physiological rules while maintaining a state-of-the-art **92.19% accuracy**.
- **Multimodal Fault Tolerance:** Integrating hemodynamic vitals (HR, BP, CO_2) ensures that the Virtual Monitor remains robust even if operating room interference corrupts the EEG signal.

Future Frontiers:

- **Decentralized AI (DeAI) & Federated Learning:** Future iterations will utilize blockchain-based zero-knowledge proofs. This allows hospitals to train the ANFIS model on local surgical data and share only the updated fuzzy weights to a global ledger, completely preserving patient privacy.
- **IoMT & Edge Computing:** Compiling the PyTorch inference engine onto localized, low-power FPGAs to ensure life-saving logic rules execute in real-time without relying on external hospital networks.

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Thank You!

Questions?

